

An Introduction to Knowledge Graphs and Retrieval-Augmented Generation

1. Graph Databases

Neo4j is a graph database that stores information as nodes and relationships instead of tables. Neo4j uses Cypher as its query language, which lets developers describe graph patterns directly in a query. The Neo4j graph database also supports a native vector index, which allows it to store both structured graph data and dense vector embeddings in a single system.

Cypher is a declarative query language developed for Neo4j. Cypher supports the MERGE operation, which creates a node or relationship only if it does not already exist, making repeated writes idempotent.

2. Large Language Models

Gemini is a family of large language models developed by Google. Gemini Flash is a faster, lighter variant of Gemini optimized for high-throughput tasks such as entity extraction. Gemini can be accessed through an API and integrated into applications using frameworks such as LangChain.

LangChain is a Python framework that manages prompt orchestration, chaining, and integration with external tools. LangChain provides a component called GraphCypherQAChain, which converts a natural language question into a Cypher query and executes it against a graph database such as Neo4j.

3. Embeddings and Entity Resolution

Sentence embeddings represent text as dense numerical vectors, allowing semantically similar text to be compared using distance metrics. The all-MiniLM-L6-v2 model is a lightweight sentence embedding model trained by the Sentence-Transformers project. The MiniLM model produces 384-dimensional vectors and runs efficiently on a standard laptop CPU.

Entity resolution is the process of identifying when two different text mentions refer to the same real-world entity. Cosine similarity is commonly used to compare embeddings during entity resolution, since it measures the angle between two vectors rather than their magnitude.

4. Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly abbreviated as RAG, combines information retrieval with text generation. A RAG system retrieves relevant context before asking a language model to generate an answer, which reduces hallucination compared to using a language model alone.

GraphRAG extends traditional RAG by retrieving information from a knowledge graph instead of, or in addition to, plain text passages. Hybrid retrieval systems combine graph-based retrieval with vector-based semantic search to improve answer accuracy.

5. Application Framework and Visualization

FastAPI is a Python web framework used to build APIs. FastAPI is built on top of Starlette and uses Pydantic for data validation. The FastAPI framework automatically generates interactive API documentation, which is useful for testing endpoints during development.

D3.js is a JavaScript library used for creating interactive data visualizations in a web browser. D3.js is commonly used to render force-directed graphs, which are well suited for displaying entities and relationships extracted from a knowledge graph.

pdfplumber is a Python library used to extract text and tables from PDF documents. pdfplumber preserves page layout information, which makes it well suited for processing technical documents that contain tables.

6. Summary

A modern knowledge graph platform typically combines several components: a document parser such as pdfplumber, a large language model such as Gemini for entity extraction, an embedding model such as all-MiniLM-L6-v2 for entity resolution, a graph database such as Neo4j for storage, a retrieval framework such as LangChain for hybrid search, a backend framework such as FastAPI for serving results, and a visualization library such as D3.js for exploration.

This document intentionally repeats several terms in slightly different forms, such as 'Neo4j', 'the Neo4j graph database', and 'Neo4j graph database', as well as 'Gemini' and 'Gemini Flash', so that it can also be used to test entity resolution logic later in the pipeline.